Appendix R: Organization Templates

Appendix R - Shop Organization Templates

This appendix provides practical templates, checklists, and forms for implementing shop organization systems in CNC manufacturing environments.

Table of Contents

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1. 5S Audit Checklists

1.1 Master 5S Audit Scorecard

```
5S AUDIT SCORECARD

Area: _________________________ Date: __________ Auditor: __________
Supervisor: _____________________ Shift: __________

SCORING: 5 = Excellent, 4 = Good, 3 = Average, 2 = Poor, 1 = Unacceptable

+----------------------------------------------------------------------+
| 1. SORT (SEIRI) - Eliminate Unnecessary Items                       |
|----------------------------------------------------------------------|
| 1.1 Only necessary items present in work area         Score: ___  |
| 1.2 No obsolete tools, fixtures, or materials         Score: ___  |
| 1.3 Red tag area properly managed                     Score: ___  |
| 1.4 Items not used in 30 days removed                 Score: ___  |
|                                            SORT TOTAL: ___ / 20      |
|----------------------------------------------------------------------|
| 2. SET IN ORDER (SEITON) - A Place for Everything                   |
|----------------------------------------------------------------------|
| 2.1 All items have designated locations               Score: ___  |
```

```
| 2.2 Locations clearly marked and labeled              Score: ___  |
| 2.3 Shadow boards used where appropriate              Score: ___  |
| 2.4 Frequently used items easily accessible           Score: ___  |
| 2.5 Tools returned to proper location after use       Score: ___  |
|                                    SET IN ORDER TOTAL: ___ / 25     |
|-------------------------------------------------------------------|
| 3. SHINE (SEISO) - Clean and Inspect                              |
|-------------------------------------------------------------------|
| 3.1 Work area clean and free of debris                Score: ___  |
| 3.2 Equipment clean and well-maintained               Score: ___  |
| 3.3 Floors clean, no oil or coolant spills            Score: ___  |
| 3.4 Cleaning supplies available and organized         Score: ___  |
| 3.5 Daily cleaning checklist completed                Score: ___  |
|                                         SHINE TOTAL: ___ / 25       |
|-------------------------------------------------------------------|
| 4. STANDARDIZE (SEIKETSU) - Create Standards                      |
|-------------------------------------------------------------------|
| 4.1 Visual standards displayed (photos, diagrams)     Score: ___  |
| 4.2 Standard locations consistent across similar areas Score: ___  |
| 4.3 Color coding used appropriately                   Score: ___  |
| 4.4 Procedures documented and accessible              Score: ___  |
|                                    STANDARDIZE TOTAL: ___ / 20      |
|-------------------------------------------------------------------|
| 5. SUSTAIN (SHITSUKE) - Maintain and Improve                      |
|-------------------------------------------------------------------|
| 5.1 Personnel follow 5S standards consistently        Score: ___  |
| 5.2 Regular audits conducted and documented           Score: ___  |
| 5.3 Improvement suggestions generated                 Score: ___  |
| 5.4 5S is part of daily routine                       Score: ___  |
|                                        SUSTAIN TOTAL: ___ / 20      |
+-------------------------------------------------------------------+

OVERALL SCORE: ___ / 110        PERCENTAGE: ___%

RATING:
90-110 points (82-100%): GREEN - Excellent
70-89 points (64-81%):   YELLOW - Satisfactory (improvement needed)
<70 points (<64%):       RED - Unsatisfactory (immediate action required)

OBSERVATIONS / STRENGTHS:
________________________________________________________________
________________________________________________________________
________________________________________________________________

DEFICIENCIES / OPPORTUNITIES:
| Item | Description | Action Required | Responsible | Due Date |
|------|-------------|-----------------|-------------|----------|
| 1    |             |                 |             |          |
| 2    |             |                 |             |          |
| 3    |             |                 |             |          |

PREVIOUS SCORE: ___  CURRENT SCORE: ___  TREND: [ ] ↑  [ ] ->  [ ] ↓

Auditor Signature: _____________________ Date: __________
```

```
Supervisor Signature: ___________________ Date: __________
Follow-up Date: __________
```

1.2 Machine Tool 5S Checklist

```
MACHINE TOOL 5S CHECKLIST

Machine #: ________ Description: ______________ Date: __________
Operator: _________________________ Auditor: __________

[ ] SORT
  [ ] Only current job tools and fixtures at machine
  [ ] No obsolete or broken tools present
  [ ] Excess materials removed
  [ ] Personal items in designated area only

[ ] SET IN ORDER
  [ ] Tool holder storage organized and labeled
  [ ] Measuring instruments in shadow board/designated location
  [ ] Work instructions at machine (current revision)
  [ ] Cleaning supplies accessible
  [ ] Safety equipment (glasses, gloves) in designated location
  [ ] Chip bins properly positioned

[ ] SHINE
  [ ] Machine wiped down (no oil, coolant, chips on exterior)
  [ ] Chuck/vise clean
  [ ] Way covers clean
  [ ] Windows/guards clean and transparent
  [ ] Control panel clean
  [ ] Floor around machine swept/clean
  [ ] Coolant tank clean (no chips floating)

[ ] STANDARDIZE
  [ ] Photo of proper setup displayed
  [ ] Daily checklist posted and current
  [ ] Color coding followed
  [ ] Labeling clear and consistent

[ ] SUSTAIN
  [ ] Operator performing daily 5S routine
  [ ] Issues identified and reported
  [ ] Previous audit items corrected

SCORE: ___ / ___ items checked = ___%

Comments: _______________________________________________________
________________________________________________________________

Corrective Actions Required:
________________________________________________________________
________________________________________________________________
```

1.3 Tool Crib 5S Checklist

```
TOOL CRIB 5S CHECKLIST

Date: __________ Auditor: __________ Tool Crib Attendant: __________

ITEM                                                          Y / N / NA
------------------------------------------------------------------------
SORT
[ ] Only active tools in crib (obsolete items removed)          ___
[ ] Damaged tools segregated for repair/disposal                ___
[ ] Excess inventory identified and reduced                     ___

SET IN ORDER
[ ] Tools organized by type/size                                ___
[ ] All locations labeled clearly                               ___
[ ] Frequently used tools easily accessible                     ___
[ ] Tool boards/racks properly utilized                         ___
[ ] Check-out system functional                                 ___
[ ] Min/Max levels established and visible                      ___

SHINE
[ ] Shelves/drawers clean and organized                         ___
[ ] Floor clean and clear                                       ___
[ ] No dust accumulation on tools                               ___
[ ] Lighting adequate                                           ___

STANDARDIZE
[ ] Tool inventory system up to date                            ___
[ ] Standard storage locations documented                       ___
[ ] Tool checkout procedures posted                             ___
[ ] Reorder procedures documented                               ___

SUSTAIN
[ ] Daily maintenance performed                                 ___
[ ] Tool inventory accuracy >95%                                ___
[ ] Previous audit findings addressed                           ___

TOTAL SCORE: ___ / ___  PERCENTAGE: ___%

Action Items: ___________________________________________________
________________________________________________________________
```

---

2. Red Tag Forms

2.1 Red Tag Template

```
+------------------------------------------------------+
|                    RED TAG                           |
|              (UNNEEDED ITEM)                         |
|------------------------------------------------------|
| Tag Number: _____________                            |
|                                                      |
```

```
| Date Tagged: _____________                          |
|                                                     |
| Tagged By: _______________________                  |
|                                                     |
| ITEM DESCRIPTION:                                   |
| ____________________________________________        |
| ____________________________________________        |
|                                                     |
| CATEGORY:                                           |
| [ ] Raw Material      [ ] Finished Goods              |
| [ ] Tool              [ ] Fixture                     |
| [ ] Equipment         [ ] Supplies                    |
| [ ] Scrap             [ ] Other: __________           |
|                                                     |
| REASON FOR RED TAG:                                 |
| [ ] Not used in past 30 days                         |
| [ ] Obsolete                                         |
| [ ] Broken/Damaged                                   |
| [ ] Excess quantity                                  |
| [ ] Unknown purpose                                  |
| [ ] Other: _________________________                 |
|                                                     |
| ORIGINAL LOCATION: ____________________             |
|                                                     |
| DISPOSITION:                                        |
| [ ] Keep (justify): ____________________             |
| [ ] Relocate to: ________________________            |
| [ ] Return to: __________________________            |
| [ ] Sell/Donate                                      |
| [ ] Scrap/Dispose                                    |
|                                                     |
| DISPOSITION DATE: _____________                     |
|                                                     |
| Approved By: _______________________                |
+-----------------------------------------------------+
```

Instructions:
1. Attach this tag to any item that is unneeded,
   obsolete, or questionable
2. Move tagged items to Red Tag Holding Area
3. Review weekly and dispose per policy
4. Items unclaimed after 30 days will be scrapped/donated

2.2 Red Tag Log

RED TAG TRACKING LOG

Department: _________________________ Month: __________

Tag #	Date	Item Description	Category	Disposition	Date Closed	Value
						\$
						\$
						\$

```
SUMMARY:
Total Items Tagged: _____
Items Kept: _____
Items Relocated: _____
Items Returned: _____
Items Sold/Donated: _____
Items Scrapped: _____

Total Value Recovered: $________
Space Freed: ________ sq ft
```

---

3. Visual Management Templates

3.1 Shadow Board Layout Template

```
SHADOW BOARD DESIGN

Area: _________________________ Date: __________

[Draw outline of each tool with tool name labeled]

Example Layout:

+---------------------------------------------------------+
|        MACHINE #5 TOOL SHADOW BOARD                     |
|---------------------------------------------------------|
|                                                         |
|   [===CALIPER===]    [==MICROMETER==]    [=DIAL=]   |
|                                          [INDICATOR]  |
|                                                         |
|   [====ALLEN KEY SET====]                              |
|                                                         |
|   [=WRENCH=]  [=WRENCH=]  [=WRENCH=]                  |
|     10mm       12mm       14mm                         |
|                                                         |
|   [===SCREWDRIVER===]  [===SCREWDRIVER===]            |
|       Flat              Phillips                       |
|                                                         |
|   [======FLASHLIGHT======]                             |
|                                                         |
+---------------------------------------------------------+

NOTES:
- Paint tool outline on board (bright color against dark background)
- Label each location clearly
- Mount at ergonomic height
- Include photo of properly organized board
- Audit weekly for compliance

Tool List:
1. 6" Digital Caliper (ID: CAL-123)
2. 0-1" Micrometer (ID: MIC-045)
```

3. Dial Indicator w/ Magnetic Base (ID: DI-012)
4. Allen Key Set 1.5-10mm (ID: AK-008)
5. Adjustable Wrench 10mm (ID: WR-101)
6. Adjustable Wrench 12mm (ID: WR-102)
7. Adjustable Wrench 14mm (ID: WR-103)
8. Screwdriver Set (ID: SD-020)
9. LED Flashlight (ID: FL-005)

Shadow Board Location: Machine #5, right side of control panel
Responsible: Operator (1st shift: John, 2nd shift: Maria)

3.2 Floor Marking Guide

FLOOR MARKING STANDARDS

COLOR CODE:

COLOR	PURPOSE	LINE TYPE
YELLOW	Aisles, Traffic Lanes	Solid 3"
WHITE	Work Areas, Equipment	Solid 2"
RED	Defect/Scrap Area	Solid 3"
BLUE	Raw Material Storage	Solid 2"
GREEN	Finished Goods	Solid 2"
ORANGE	WIP / In-Process Inspection	Dashed 2"
BLACK/YELLOW	Safety Hazard Areas	Striped 3"

MARKING SPECIFICATIONS:

Aisles (Yellow):
- Main aisles: 3" solid yellow lines on both sides
- Minimum aisle width: 4 feet (forklifts), 3 feet (foot traffic)
- Intersections: Striped for visibility
- Arrows indicate direction (one-way aisles)

Work Areas (White):
- Define machine/workstation footprint
- Include operator work zone
- Label area (e.g., "MILL #3")

Storage Areas:
- Blue = Raw Material
- Green = Finished Goods
- Orange = WIP
- Label with part number range or customer

Safety Areas (Black/Yellow Stripe):
- Fire extinguisher zones (3' radius)
- Emergency exit paths
- Electrical panel clearance (3' x 3')
- No storage zones

MAINTENANCE:

- Inspect monthly
- Re-tape worn areas quarterly
- Use industrial floor tape (3M or equivalent)
- Clean floor before application

3.3 Visual Performance Board Template

```
+--------------------------------------------------------------+
|              DAILY MANAGEMENT BOARD                          |
|                  Department: CNC Machining                   |
|                  Date: __________                            |
|--------------------------------------------------------------|
|                                                              |
|  SAFETY                                                      |
|  Days Since Last Incident: [____]  Goal: 365                |
|  Near Misses This Week: [____]                              |
|                                                              |
|--------------------------------------------------------------|
|  QUALITY                          TARGET    ACTUAL   R/Y/G  |
|  First Pass Yield                  98%      ___%            |
|  Customer Rejects                   0       ___             |
|  Internal Scrap Rate               <1%      ___%            |
|                                                              |
|--------------------------------------------------------------|
|  DELIVERY                         TARGET    ACTUAL   R/Y/G  |
|  On-Time Delivery                  95%      ___%            |
|  Jobs Completed Today              12       ___             |
|  Schedule Adherence                90%      ___%            |
|                                                              |
|--------------------------------------------------------------|
|  PRODUCTIVITY                     TARGET    ACTUAL   R/Y/G  |
|  Overall Equipment Effectiveness   75%      ___%            |
|  Setup Time (average)              20 min   ___ min         |
|  Parts per Hour                    8.5      ___             |
|                                                              |
|--------------------------------------------------------------|
|  CONTINUOUS IMPROVEMENT                                      |
|  Open Issues: ___                                            |
|  Issues Closed This Week: ___                                |
|  Kaizen Ideas Submitted: ___                                 |
|                                                              |
|--------------------------------------------------------------|
|  TODAY'S PRIORITIES:                                         |
|  1. ____________________________________________             |
|  2. ____________________________________________             |
|  3. ____________________________________________             |
|                                                              |
|  ISSUES / BLOCKERS:                                          |
|  ________________________________________________            |
|  ________________________________________________            |
|                                                              |
+--------------------------------------------------------------+
```

Color Code: Green = On Target, Yellow = Warning, Red = Action Required

Updated by: _____________ Time: _______

---

4. Tool Management Forms

4.1 Tool Checkout/Return Log

TOOL CHECKOUT LOG

Tool Crib: _________________________

Date	Time Out	Tool ID	Description	Employee Name	Employee ID	Purpose	Returned Date/Time	Condition (R/D/M)	Initials

Condition Codes:
R = Returned OK
D = Damaged (describe below)
M = Missing/Not Returned

Notes: ____
____

4.2 Tool Inventory Control Card

TOOL INVENTORY CONTROL CARD

Tool Description: ___________________________
Part Number: _____________ Manufacturer: ______________
Unit Cost: $________

INVENTORY LEVELS:
Minimum: _____ Maximum: _____ Reorder Point: _____

CURRENT INVENTORY:

Location	Quantity	Condition
Tool Crib	_____	[ ] Good [ ] Fair [ ] Poor
Machine #1	_____	[ ] Good [ ] Fair [ ] Poor
Machine #2	_____	[ ] Good [ ] Fair [ ] Poor
Maintenance	_____	[ ] Good [ ] Fair [ ] Poor
TOTAL	_____	

TRANSACTION LOG:

Date	Type (In/Out)	Qty	Balance	Initials

REORDER INFORMATION:
Supplier: _______________________________
Lead Time: _______ days

```
Order Quantity: _______
Last Order Date: _______
Next Review Date: _______
```

4.3 Cutting Tool Life Tracking

```
CUTTING TOOL LIFE LOG

Tool Description: ___________________________
Tool ID: _____________ Purchase Date: __________
Cost: $________ Expected Life: _______ parts

USAGE LOG:
| Date | Part # | Qty | Running Total | Condition | Action | Operator |
|------|--------|-----|---------------|-----------|--------|----------|
|      |        |     |               | G/F/P     |        |          |

Condition: G=Good, F=Fair (watch), P=Poor (replace)
Action: C=Continue, R=Regrind, X=Replace

LIFECYCLE SUMMARY:
Total Parts Machined: _______
Cost per Part: $_______ (Tool Cost / Total Parts)
Actual vs. Expected Life: _______%

Replacement Notes: __________________________________________
_________________________________________________________________

Disposal Date: __________ Disposed By: __________
```

---

5. Production Planning Templates

5.1 Weekly Production Schedule Board

```
WEEKLY PRODUCTION SCHEDULE

Week of: __________ to __________

MACHINE: _____________ Operator: _____________

+------+----------+----------+----------+----------+----------+
| TIME |  MONDAY  | TUESDAY  | WEDNESDAY| THURSDAY |  FRIDAY  |
|------+----------+----------+----------+----------+----------|
| 6-8  |          |          |          |          |          |
|      | WO:      | WO:      | WO:      | WO:      | WO:      |
|      | Part:    | Part:    | Part:    | Part:    | Part:    |
|      | Qty:     | Qty:     | Qty:     | Qty:     | Qty:     |
|------+----------+----------+----------+----------+----------|
| 8-10 |          |          |          |          |          |
|      |          |          |          |          |          |
|------+----------+----------+----------+----------+----------|
|10-12 |          |          |          |          |          |
|      |          |          |          |          |          |
```

```
|------+----------+----------+----------+----------+----------|
|12-2  |          |          |          |          |          |
|      |          |          |          |          |          |
|------+----------+----------+----------+----------+----------|
| 2-4  |          |          |          |          |          |
|      |          |          |          |          |          |
+------+----------+----------+----------+----------+----------+

COLOR CODE:
 Green = On Schedule
 Yellow = Delayed (< 2 hours)
 Red = Critical (> 2 hours behind)

Notes / Issues:
________________________________________________________________
________________________________________________________________
```

5.2 Job Traveler Template

```
+--------------------------------------------------------------+
|                       JOB TRAVELER                           |
|--------------------------------------------------------------|
| Work Order #: __________  Rev: ___  Date Released: ______   |
| Part Number: ____________  Part Name: __________________    |
| Customer: __________________  PO #: __________              |
| Quantity Ordered: _____  Due Date: __________               |
| Material: ______________  Spec: __________                  |
| Heat/Lot #: __________  Certificate Req'd: [ ] Yes [ ] No    |
|--------------------------------------------------------------|
| OPERATION ROUTING:                                           |
|                                                              |
| Op 10: Raw Material Prep                                    |
|   Machine/Area: Saw        Setup: 10 min  Run: 5 min/pc   |
|   Start: ______ Complete: ______ Qty: ____ OK: ___ Scrap: ___|
|   Operator: ____________  Inspector: ____________           |
|                                                              |
| Op 20: CNC Milling - Op 1                                   |
|   Machine: Mill #3         Setup: 45 min  Run: 12 min/pc  |
|   Program: P-1234-01       Tools: Per setup sheet           |
|   Start: ______ Complete: ______ Qty: ____ OK: ___ Scrap: ___|
|   First-Off Inspection: [ ] Approved  Inspector: __________  |
|   Operator: ____________                                     |
|                                                              |
| Op 30: CNC Turning                                          |
|   Machine: Lathe #2        Setup: 30 min  Run: 8 min/pc   |
|   Program: P-1234-02       Tools: Per setup sheet           |
|   Start: ______ Complete: ______ Qty: ____ OK: ___ Scrap: ___|
|   Operator: ____________                                     |
|                                                              |
| Op 40: Deburr                                               |
|   Machine/Area: Bench      Run: 5 min/pc                    |
|   Start: ______ Complete: ______ Qty: ____ OK: ___ Scrap: ___|
|   Operator: ____________                                     |
|                                                              |
```

```
| Op 50: Final Inspection                                   |
|   Inspection Plan: IP-1234  Sample Size: 100%            |
|   Start: ______ Complete: ______ Qty: ____ OK: ___ Reject: ___|
|   Inspector: ____________                                 |
|   Disposition: [ ] Ship  [ ] Hold  [ ] Scrap  [ ] Rework        |
|                                                            |
|-----------------------------------------------------------|
| SPECIAL INSTRUCTIONS:                                      |
| _________________________________________________________|
|                                                            |
| QUALITY RECORDS ATTACHED:                                  |
| [ ] First Article Inspection  [ ] Material Cert              |
| [ ] Inspection Report         [ ] Other: __________          |
|                                                            |
| FINAL DISPOSITION:                                         |
| Quantity Shipped: _____  Date: __________                  |
| Shipped By: ____________  Packing Slip #: __________       |
+-----------------------------------------------------------+
```

---

6. Inventory Management Forms

6.1 Cycle Count Sheet

```
CYCLE COUNT SHEET

Date: __________ Counter: __________ Verified By: __________

Location/Bin: __________

| Item | Description | Part # | Unit | Expected | Actual | Variance | Reason Code |
|------|-------------|--------|------|----------|--------|----------|-------------|
|      |             |        |      |          |        |          |             |

Reason Codes:
S = Shrinkage/Loss
M = Misplaced/Wrong Location
T = Transaction Error (not recorded)
D = Damaged/Scrapped (not recorded)
O = Other (explain in notes)

ACCURACY:
Items Counted: _____
Items Accurate: _____
Accuracy Rate: _____%

Variances:
Overage: _____ items  Value: $_______
Shortage: _____ items  Value: $_______

Action Required: [ ] Adjust Inventory  [ ] Investigate  [ ] Recount

Notes: __________________________________________________________
________________________________________________________________
```

```
Supervisor Approval: _____________________ Date: __________
```

6.2 Material Kanban Card

```
+--------------------------------------------------+
|              KANBAN CARD                         |
|            (Production)                          |
|--------------------------------------------------|
|                                                  |
| Part Number: ____________________                |
|                                                  |
| Part Description: ____________________           |
|                                                  |
| Container Quantity: ________                     |
|                                                  |
| Container Type: [ ] Bin [ ] Pallet [ ] Tote        |
|                                                  |
| Storage Location: ____________________           |
|                                                  |
| Reorder Point: ________ containers               |
|                                                  |
| Lead Time: ________ days/hours                   |
|                                                  |
| Supplier (if external): ____________________     |
|                                                  |
| Production Area (if internal): _____________     |
|                                                  |
|--------------------------------------------------|
| INSTRUCTIONS:                                    |
| 1. When container is empty, return card to       |
|    Kanban board                                  |
| 2. Card triggers replenishment order              |
| 3. Do not produce/order without Kanban card      |
+--------------------------------------------------+

Card # ___ of ___ (Total cards in system)
```

---

7. Preventive Maintenance Templates

7.1 Daily Machine Checklist

```
DAILY MACHINE INSPECTION CHECKLIST

Machine: _____________ Machine #: _____ Date: __________
Operator: _________________ Shift: [ ] 1st [ ] 2nd [ ] 3rd

START OF SHIFT (Complete before operating):
[ ] Machine is clean (chips removed, wiped down)
[ ] Chuck/vise clean and undamaged
[ ] Coolant level adequate
[ ] Oil levels checked (way lube, hydraulic)
[ ] No unusual noises during startup
```

```
[ ] Emergency stop functional
[ ] Guards and safety devices in place
[ ] Control panel operational
[ ] All axes move freely without binding
[ ] Spindle runs smoothly (no vibration)

DURING SHIFT (Monitor continuously):
[ ] Coolant flow adequate
[ ] Chips evacuating properly
[ ] No unusual vibration or noise
[ ] Temperatures normal
[ ] Accuracy maintained

END OF SHIFT (Complete before leaving):
[ ] Clean machine (remove chips and debris)
[ ] Check and top off coolant
[ ] Wipe down exterior surfaces
[ ] Return tools to proper location
[ ] Document any issues below
[ ] Complete machine log

ISSUES / CONCERNS:
[ ] None - Machine operating normally

Issue: __________________________________________________________
Severity: [ ] Minor (note only)  [ ] Moderate (repair soon)  [ ] Critical (stop work)
Action Taken: ___________________________________________________
Maintenance Notified: [ ] Yes (Name: ___________) [ ] No

Operator Signature: _____________________ Time: _______
```

7.2 Preventive Maintenance Schedule

```
PREVENTIVE MAINTENANCE SCHEDULE

Equipment: _________________ Equipment ID: _____

DAILY (Operator-Performed):
[ ] Visual inspection
[ ] Cleaning and chip removal
[ ] Fluid level checks
[ ] Lubrication (if required)
Last completed: ________ By: ________

WEEKLY:
[ ] Detailed cleaning
[ ] Check way cover condition
[ ] Inspect hoses and fittings for leaks
[ ] Test emergency stops
Last completed: ________ By: ________

MONTHLY:
[ ] Check and adjust coolant concentration
[ ] Inspect spindle for runout
```

```
[ ] Check/clean filters
[ ] Lubricate as per manufacturer schedule
[ ] Check tool holder retention
Last completed: ________ By: ________

QUARTERLY:
[ ] Change hydraulic oil filter
[ ] Inspect drive belts
[ ] Check backlash in axes
[ ] Inspect ball screws
[ ] Calibrate level (if critical)
Last completed: ________ By: ________

ANNUALLY:
[ ] Full machine calibration (laser)
[ ] Replace way lube oil
[ ] Replace coolant
[ ] Electrical inspection
[ ] Ball screw inspection/regrease
[ ] Spindle inspection
Last completed: ________ By: ________

NEXT SCHEDULED MAINTENANCE:
Type: ____________  Due Date: ________  Assigned To: ________
```

---

8. Standard Work Templates

8.1 Standardized Work Chart

```
STANDARDIZED WORK CHART

Process: _________________________ Part #: _____________
Work Station: ____________________ Rev: ___ Date: _________

TAKT TIME: _______ seconds (Available time / Customer demand)
CYCLE TIME: _______ seconds (Total time for one complete cycle)

WORK SEQUENCE:

+-------------------------------------------------------------+
|         [Include simple floor plan/layout sketch]           |
|                                                             |
|    [Show workstation, machines, material locations,        |
|     movement paths with arrows and numbered steps]         |
+-------------------------------------------------------------+

WORK ELEMENT BREAKDOWN:
```

Step	Description	Time(sec)	Walking	Machine	Manual	Wait	Check
1				[check]			
2			[check]				
3					[check]		

```
| 4    |              |           |         |         |        |  [check]   |        |
| 5    |              |           |         |         |        |      |   [check]   |
|      | TOTAL        |           |         |         |        |      |       |

STANDARD WIP (Work in Progress): _____ pieces

QUALITY CHECKS:
Critical Dimension: ___________  Frequency: Every ____ pieces
Visual Inspection: ___________  Frequency: Every ____ pieces

SAFETY:
[ ] Safety glasses required
[ ] Hearing protection required
[ ] Gloves required (specify type): __________
[ ] Other PPE: __________

NOTES / SPECIAL INSTRUCTIONS:
----------------------------------------------------------------
----------------------------------------------------------------

Prepared by: _____________ Approved by: _____________ Date: ____
```

8.2 Job Instruction Sheet (Training)

```
JOB INSTRUCTION SHEET

Job: _________________________ Department: _____________
Trainer: _____________________ Date Prepared: __________

TOOLS AND MATERIALS NEEDED:
- ___________________________
- ___________________________
- ___________________________

SAFETY PRECAUTIONS:
- ___________________________
- ___________________________

IMPORTANT POINTS TO REMEMBER:
- ___________________________
- ___________________________

INSTRUCTION BREAKDOWN:

+---------------------------------------------------------------+
| STEP | KEY POINT (What to do) | REASON (Why it matters)   |
|------+------------------------+--------------------------|
|  1   |                        |                          |
|      |                        |                          |
|------+------------------------+--------------------------|
|  2   |                        |                          |
|      |                        |                          |
|------+------------------------+--------------------------|
|  3   |                        |                          |
```

```
|      |                        |                          |
|------+------------------------+--------------------------|
|  4   |                        |                          |
|      |                        |                          |
+--------------------------------------------------------------+

TRAINING CHECKLIST:
[ ] Trainee observed complete job
[ ] Trainer explained each step
[ ] Trainee performed job under supervision
[ ] Trainee performed job independently
[ ] Trainer verified competency

Trainee: _________________ Trainer: _________________ Date: ____
Supervisor Approval: _____________________ Date: __________
```

---

9. Performance Metrics Dashboards

9.1 OEE (Overall Equipment Effectiveness) Tracking

```
OEE TRACKING SHEET

Machine: _____________ Week of: __________

DAILY OEE CALCULATION:

DAY: MONDAY      Date: __________
-----------------------------------------------------
Scheduled Production Time:           480 minutes
Planned Downtime (breaks, meetings): -30 minutes
-----------------------------------------------------
AVAILABLE TIME:                      450 minutes

Actual Downtime:
  - Setup/Changeover:     ___ min
  - Breakdowns:           ___ min
  - Material shortage:    ___ min
  - Other:                ___ min
  Total Downtime:         ___ minutes
-----------------------------------------------------
OPERATING TIME:                      ___ minutes
AVAILABILITY = Operating Time / Available Time = ___%

Ideal Cycle Time:               ___ seconds/piece
Total Pieces Produced:          ___ pieces
-----------------------------------------------------
PERFORMANCE = (Total Pieces x Ideal Time) / Operating Time = ___%

Total Pieces Produced:          ___ pieces
Good Pieces:                    ___ pieces
Defective Pieces:               ___ pieces
-----------------------------------------------------
QUALITY = Good Pieces / Total Pieces = ___%
```

```
OEE = AVAILABILITY x PERFORMANCE x QUALITY = ___%

WEEKLY SUMMARY:
```

Day	Availability	Performance	Quality	OEE
Mon	%	%	%	%
Tue	%	%	%	%
Wed	%	%	%	%
Thu	%	%	%	%
Fri	%	%	%	%
AVG	%	%	%	%

```
TARGET OEE: 75%     WORLD CLASS: 85%+

TOP LOSSES THIS WEEK:
1. ______________________  Impact: ___ minutes
2. ______________________  Impact: ___ minutes
3. ______________________  Impact: ___ minutes

IMPROVEMENT ACTIONS:
----------------------------------------------------------------
----------------------------------------------------------------
```

---

10. Facility Layout Planning

10.1 Spaghetti Diagram Template

```
SPAGHETTI DIAGRAM
(Material/Operator Movement Analysis)

Process: _________________________ Date: __________
Observer: ________________________ Duration: _______ hours

INSTRUCTIONS:
1. Draw facility layout to scale (walls, machines, storage)
2. Track actual movement of operator or material
3. Draw lines following each movement
4. Record distance and time for each movement
5. Analyze for waste and improvement opportunities

[LAYOUT SKETCH AREA - Draw walls, machines, workstations]

+----------------------------------------------------------+
|                                                          |
|                                                          |
|                                                          |
|                                                          |
|                                                          |
|                                                          |
|                                                          |
```

```
|                                                         |
|                                                         |
+---------------------------------------------------------+

MOVEMENT LOG:

| # | From | To | Purpose | Distance(ft) | Time(min) | Value-Added? |
|---|------|----|---------|--------------|-----------|--------------|
| 1 |      |    |         |              |           | Y / N        |
| 2 |      |    |         |              |           | Y / N        |

SUMMARY:
Total Distance Traveled: ________ feet
Total Time Spent Moving: ________ minutes
Value-Added Movements: ________
Non-Value-Added Movements: ________

IMPROVEMENT OPPORTUNITIES:
1. ________________________________________________________
2. ________________________________________________________
3. ________________________________________________________

ESTIMATED SAVINGS:
Distance Reduction: ________ feet/day
Time Reduction: ________ minutes/day
Annual Savings: $________ (Time x Labor Rate x 250 days)
```

Document Control Information

```
Appendix R - Shop Organization Templates
Revision: A
Date: __________
Approved by: ___________________________

These templates are provided as guidance and should be customized to meet
specific organizational needs and processes.

All forms should be reviewed annually and updated as needed to maintain
alignment with current practices.
```

Implementation Notes

1. **Customize templates** for your specific shop environment and needs
2. **Digitize forms** where possible (tablets, QMS software) for easier data collection
3. **Train personnel** on proper use of all forms and checklists
4. **Audit compliance** regularly to ensure templates are being used correctly
5. **Revise as needed** based on user feedback and process changes
6. **Maintain completed forms** per record retention requirements
7. **Use visual management** principles (color coding, graphics) to make forms intuitive